Belief-Constrained Inference: Correcting Visual Premises at Inference Time for More Reliable Vision-Language Models

Anonymous CVPR submission

Paper ID *****

Abstract

Vision-Language Models (VLMs) frequently err not because they reason incorrectly, but because they reason from incorrect visual premises*—false perceptual beliefs about what an image contains. We introduce* ***Belief-Constrained Inference*** *(*BCI*), an inference-time framework that externalizes, verifies, and corrects a model's intermediate visual beliefs before final answer generation, without modifying model weights.*

BCI *prompts the VLM to emit a set of atomic, falsifiable visual claims as an explicit reasoning substrate. These claims are verified against structured visual evidence (scene graphs) and, if contradicted, are replaced with ground-truth facts prior to constrained re-reasoning. On GQA (n=500),* BCI *with full premise replacement (**E1**) raises accuracy from 59.2% to 77.2% (+18.0 pp; McNemar p=1.5×10^{-14})—a causal upper bound establishing premise correction as a highly effective intervention axis. Scaled to n=2000, E1 achieves +16.2 pp (63.55% to 79.75%; p=1.78×10^{-48}) with a flip-to-correct rate of 21.2%, confirming consistent gains at scale. A practical confidence-gated variant (**E12b**, n=2000) achieves +3.45 pp (p=1.3×10^{-5}) while intervening on only 59.85% of samples, with multi-seed stability of* $66.18\% \pm 0.14$ *pp. Critically, targeted correction outperforms random belief deletion by +35 pp, confirming that gains are claim-specific rather than a generic perturbation effect.* BCI *also outperforms matched lightweight baselines including self-correction (49.2%) and chain-of-verification (57.2%), establishing it as a strong and interpretable inference-time reliability mechanism for VLMs.*

1. Introduction

Vision-Language Models (VLMs) have achieved remarkable performance across a wide range of perception and reasoning tasks [2, 5, 13]. Yet they remain prone to systematic failures that are difficult to diagnose or correct post-hoc. When a VLM answers a question about an image incorrectly, the failure may stem from two distinct sources: (1) a *reasoning error*—incorrect inference from accurate perceptual inputs; or (2) a *premise error*—reasoning from false beliefs about what the image contains. Standard evaluation does not distinguish these two failure modes, and standard training procedures do not explicitly enforce that intermediate beliefs be grounded in visual evidence.

This paper asks a deceptively simple question: *what fraction of VLM errors are caused by false visual premises, and can correcting them at inference time recover the correct answer?*

The Epistemic Failure Hypothesis. We hypothesize that a substantial portion of VLM failures are *epistemic* in nature—the model's answer is wrong because it begins reasoning from an incorrect perception of the scene, not because its reasoning chain is flawed. If this holds, then selectively correcting false beliefs at inference time, without any weight updates, should flip wrong answers to correct. This is a falsifiable, causal claim: under a random-ablation control (removing beliefs without targeted correction), we would expect no such systematic improvement.

BCI: Belief-Constrained Inference. We propose a structured inference-time loop operating in five stages:

1. **Belief Externalization.** Before producing a final answer, the VLM emits 5–10 atomic, falsifiable visual claims about the image.
2. **Claim Extraction.** Claims are parsed into typed atomic predicates (object, attribute, relation, action, count).
3. **Claim Verification.** Each claim is checked against structured visual evidence (GQA scene graphs), yielding SUPPORTED, CONTRADICTED, or UNCERTAIN.
4. **Belief Revision.** Contradicted claims are replaced with ground-truth facts (E1), removed (E3), or selectively corrected via a confidence gate (E12b).
5. **Constrained Re-reasoning.** The model re-answers using only the revised belief set.

Key Findings. On GQA under Qwen2.5-VL-7B [2], 33.3% of all incorrect answers involve at least one contradicted belief, rising to 50% for logical/relational questions. Full premise replacement (E1) yields +18.0 pp at n=500 (p=1.5×10^{-14}) and +16.2 pp at n=2000 (p=1.78×10^{-48}). Across both scales, flip-to-correct rates exceed 21%, while flip-to-wrong rates remain below 6%. Targeted correction outperforms random belief deletion by +35 pp, confirming that the effect is claim-specific and causal. A practical, compute-bounded gated policy (E12b) achieves +3.45 pp at n=2000 (p=1.3×10^{-5}), demonstrating that the causal signal is harvestable under realistic constraints. Against matched lightweight baselines, BCI-E1 leads the competitor matrix by 17+ pp, outperforming Resample-Lite (60.6%), CoVe-Lite (57.2%), and Self-Correct (49.2%).

Contributions. We summarize our contributions as:

- We introduce BCI, the first systematic framework for *premise-level* inference-time error correction in VLMs, explicitly distinguishing epistemic (premise) from logical (reasoning) failures.
- We provide causal evidence—via targeted *vs*. random ablation—that premise correction is a specific and effective intervention axis, with a +35 pp gap over random belief deletion at matched removal budget.
- We define an intervention policy family spanning from an oracle upper bound (E1: +18.0 pp, +16.2 pp at n=2000) to a practical confidence-gated variant (E12b: +3.45 pp at n=2000, p=1.3×10^{-5}) suitable for deployment.
- We characterize verifier behavior across three profiles and four noise levels (0%–20%), showing that the pipeline degrades gracefully and that the 33.3% premise-error rate is a conservative lower bound.
- We release a reproducible codebase with versioned verifier profiles, YAML-driven experiment configs, and full run manifests (git hash, code hash, config hash, random seed).

2. Related Work

VLM Hallucination. Hallucination in large vision-language models—generating content inconsistent with the visual input—has been extensively documented [7, 12]. Existing work identifies hallucinations in object existence, attributes, spatial relations, and counts. Most prior work treats hallucination as a generation artifact and addresses it via training-time regularization or decoding modifications [9, 11]. BCI instead reframes the problem as an *epistemic* failure amenable to inference-time structural correction, without changing model parameters.

Post-Hoc Verification and Correction. Woodpecker [16] is the closest prior work: it detects and corrects hallucinations post-hoc by querying specialized models and retrieving image evidence. BCI differs in two key respects: (1) it corrects beliefs *before* the final answer is generated, closing the reasoning loop at the premise level; and (2) it quantifies the causal contribution of false premises via controlled ablation. Chain-of-Verification (CoVe) [4] generates targeted re-check questions after an initial response and revises accordingly; this is a complementary self-refinement approach, but it does not decompose failures into premise *vs*. reasoning types. Our matched in-house evaluation of a CoVe-lite baseline confirms that BCI outperforms this strategy on GQA.

Inference-Time Intervention. A growing literature explores inference-time methods for improving LLM and VLM reliability. Self-consistency [15] aggregates multiple samples; resampling-based methods [3] filter by answer frequency. Visual Contrastive Decoding [11] modifies logit distributions to suppress hallucination. These approaches operate at the *output distribution* level, whereas BCI intervenes at the *intermediate belief* level, offering interpretability and fine-grained error attribution.

Scene Graph Grounding. GQA [10] provides structured scene-graph annotations that make formal claim verification tractable. We use these as a deterministic verifier in Phase 1 experiments, deliberately separating verifier design from the core intervention hypothesis. Extending BCI to neural or learned verifiers is a natural next step.

Error Decomposition. Prior work on VQA error analysis [1, 6] typically operates at the answer level. Our premise/reasoning decomposition provides a finer-grained taxonomy more suitable for targeted intervention design.

3. Method

3.1. Problem Formulation

Let (I, Q, A^*) denote an image, a natural-language question, and the ground-truth answer. A VLM $\mathcal{M}$ produces $\hat{A} = \mathcal{M}(I, Q)$. Standard evaluation assesses $\hat{A} = A^*$ but does not attribute errors. We decompose errors into two types:

- **Premise Error (PE):** $\hat{A} \neq A^*$ *and* at least one extracted belief $b_i \in \mathcal{B}$ is verifiably false with respect to I.
- **Reasoning Error (RE):** $\hat{A} \neq A^*$ *and* no extracted belief is verifiably false.

BCI aims to (a) quantify the PE rate and (b) provide inference-time interventions that convert PE cases to correct answers.

3.2. The BCI Pipeline

Figure 1 illustrates the full five-stage pipeline.

Stage 1: Belief Externalization. Before producing a final answer, $\mathcal{M}$ is instructed to emit a set of 5–10 atomic, grounded, falsifiable visual beliefs: $\mathcal{B} = \{b_1, b_2, \ldots, b_k\}$, $k \in [5, 10]$. Each b_i is a declarative sentence asserting a visual fact (*e.g.*, "the cup is red," "the person is left of the table").

Stage 2: Claim Extraction. Beliefs are parsed into typed atomic claims (e, t, v) where e is the entity, $t \in \{\text{object}, \text{attribute}, \text{relation}, \text{action}, \text{count}\}$ is the claim type, and v is the asserted value.

Stage 3: Claim Verification. Each claim is verified against a structured evidence source. The verifier $\mathcal{V}$ assigns:

$$\ell_i = \mathcal{V}(b_i, \mathcal{G}_I) \in \{\textsc{Supported}, \textsc{Contradicted}, \textsc{Uncertain}\} \tag{1}$$

We define three versioned verifier profiles—`strict`, `balanced`, `high_recall`—parameterized by attribute similarity thresholds $\theta_{\text{attr}} \in \{0.7, 0.8, 0.9\}$ and spatial tolerances $\delta \in \{0.10, 0.15, 0.20\}$.

Stage 4: Intervention Policy. Given labeled beliefs, a policy π maps $\mathcal{B}$ to a corrected set $\mathcal{B}'$. We study three families:

- **E1 – Replace All** (causal upper bound): all contradicted claims are replaced with ground-truth facts extracted from the scene graph.
- **E3 – Remove Only** (conservative): contradicted claims are deleted without replacement, testing whether removal alone suffices.
- **E12b – Confidence-Gated** (practical): a scalar intervention score $s \in [0, 1]$ is computed from verifier confidence and claim-type reliability. Replacement is applied only when $s \geq \tau$ (with τ=0.18 after threshold sweep).

Stage 5: Constrained Re-reasoning. The model re-answers using $\mathcal{B}'$ as an explicit fact prefix:

$$\hat{A}' = \mathcal{M}(I, Q, \mathcal{B}') \tag{2}$$

3.3. Evaluation Protocol

Pairs $(\hat{A}, \hat{A}')$ are compared against A^*. We report:

- **F↑ (Flip-to-Correct):** fraction where $\hat{A} \neq A^*$ and $\hat{A}' = A^*$.
- **F↓ (Flip-to-Wrong):** fraction where $\hat{A} = A^*$ and $\hat{A}' \neq A^*$.
- **ΔAcc:** net accuracy difference.
- **Statistical significance:** McNemar's exact test on paired outcomes.

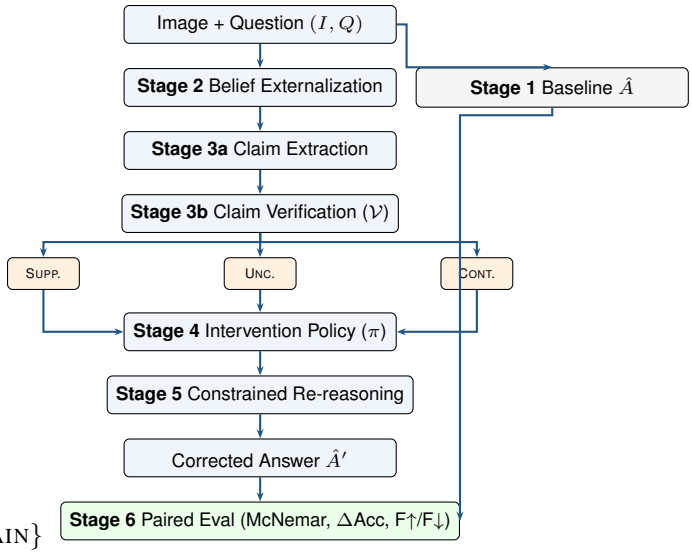

Figure 1. The **BCI pipeline**. Belief externalization and structured verification enable targeted, claim-level correction before constrained re-reasoning. Both baseline and corrected answers feed into a paired statistical evaluation.

4. Experimental Setup

Dataset. We evaluate on **GQA** [10] (balanced validation split). GQA provides compositional visual questions with associated scene-graph annotations, enabling automated belief verification. Questions span five structural types: `query`, `verify`, `choose`, `logical`, and `compare`. We sample n=500 (main experiments) and n=2000 (large-scale experiments) using seed 42, restricted to questions with matching scene-graph image IDs. The n=2000 split covers 887 query-type, 854 verify-type, 202 other-type, and 57 logical-type questions, reflecting the natural benchmark distribution.

Models. We use **Qwen2.5-VL-7B-Instruct** [2] as our primary VLM. Early experiments with LLaVA-1.5-7B [13] produced insufficient belief externalization quality (avg. 2.17 beliefs/question vs. 4.96 for Qwen), making premise-error detection unreliable (Tab. 7). All reported results use Qwen unless otherwise noted.

Verifier. Claims are verified deterministically against GQA scene graphs. We define three versioned *verifier profiles* (Eq. (1)): `strict` (attribute threshold θ=0.9, spatial tolerance δ=0.10), `balanced` (θ=0.8, δ=0.15), and `high_recall` (θ=0.7, δ=0.20). Main experiments use `balanced`; profile sensitivity and noise robustness are evaluated via a dedicated verifier audit (E10/E11) in Sec. 6.1. Every run is logged with a reproducibility manifest (git hash, code hash, config hash, random seed).

Baselines. We compare against matched in-house baselines using the same model, split, and seed:

- **Vanilla:** direct VLM inference, no intervention.
- **Self-Correct:** re-prompt the model to revise its own answer without external feedback [14].
- **CoVe-Lite:** lightweight chain-of-verification with targeted re-check questions [4].
- **Resample-Lite:** sample three independent answers, take majority vote.
- **BCI-E3:** BCI with remove-only policy (conservative ablation).
- **BCI-E1:** BCI with full ground-truth replacement (proposed upper bound).
- **BCI-E12b:** BCI with confidence-gated replacement (practical variant; threshold τ=0.18 selected via held-out sweep).

Inference Configuration. All VLM calls use temperature 0.0 and a maximum of 768 new tokens. Experiments are run on 4× NVIDIA RTX 6000 Ada (49 GB each). BCI adds two additional inference passes per sample (externalization and re-reasoning) relative to the Vanilla baseline. Wall-clock runtimes for the n=2000 runs are 15,629 s for E1 and 15,980 s for E12b (approximately 7.8 s and 8.0 s per sample, respectively). The cost-efficiency ratio (accuracy gain per second per sample) is 2.07 for E1 and 0.43 for E12b, reflecting the trade-off between intervention strength and compute.

5. Results

5.1. Error Decomposition

Table 1 and Fig. 2 report the premise/reasoning breakdown for incorrect answers under Qwen2.5-VL on GQA (n=500). Of 204 incorrect answers, **68 (33.3%)** involve at least one verifiably contradicted belief, constituting *premise errors*. The remaining 136 (66.7%) are reasoning errors. Mean belief coverage over scene-graph facts is 0.49 (see Fig. 9), so the 33.3% figure is a *conservative lower bound*. Qwen2.5-VL externalizes an average of 4.96 beliefs per question, enabling substantially higher premise-error detection than LLaVA-1.5 (2.17 beliefs; 18.8% PE rate).

5.2. Intervention Results

Table 2 summarizes all intervention methods at both scales. **BCI-E1** achieves the largest gain at n=500: +18.0 pp (59.2% → 77.2%), with 117 samples flipped to correct and only 27 regressed (McNemar $p=1.5\times10^{-14}$). Figure 3 shows that among the 68 premise-error cases, correcting false beliefs directly recovers the correct answer in 60.3% of them. At n=2000, E1 achieves +16.2 pp (63.55% → 79.75%; $p=1.78\times10^{-48}$) with a flip-to-correct rate of 21.2%, confirming that the causal signal scales robustly.

Table 1. Error decomposition by question type (Qwen2.5-VL, GQA, n=500). Premise errors are most prevalent in logical/relational questions.

Question Type	Premise Error	Reasoning Error
logical	50.0%	50.0%
compare	40.0%	60.0%
query	32.9%	67.1%
verify	30.4%	69.6%
choose	14.3%	85.7%
Overall	**33.3%**	**66.7%**

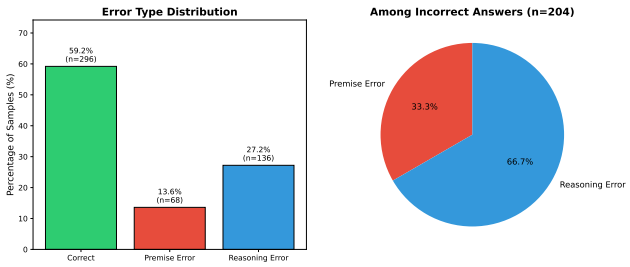

Figure 2. Error type distribution for Qwen2.5-VL on GQA (n=500). Left: overall breakdown (59.2% correct, 13.6% PE, 27.2% RE). Right: among incorrect answers, 33.3% are premise errors—a conservative lower bound given 49% mean claim coverage.

BCI-E3 shows near-zero net gain at n=500 (−0.6 pp, p=0.84), establishing that *removal alone is insufficient*—replacing false beliefs with ground-truth facts is the operative mechanism.

BCI-E12b achieves +3.45 pp at n=2000 ($p=1.3\times10^{-5}$) while intervening on 59.85% of samples. Multi-seed evaluation (seeds 42, 123, 456) yields 66.18% ± 0.14 pp, confirming numerical robustness.

Table 2. Main intervention results on GQA. BCI-E1 is the causal upper bound; E12b is the practical bounded-intervention policy. McNemar exact p-values are against the Vanilla baseline. F↑: flip-to-correct rate; F↓: flip-to-wrong rate.

Method	n	Acc.	ΔAcc.	F↑	F↓
Vanilla	500	59.2%	—	—	—
Self-Correct	500	49.2%	−10.0 pp	—	—
CoVe-Lite	500	57.2%	−2.0 pp	—	—
Resample-Lite	500	60.6%	+1.4 pp	—	—
BCI-E3	500	58.6%	−0.6 pp	9.4%	10.0%
BCI-E1	500	**77.2%**	**+18.0 pp**	**23.4%**	5.4%
Vanilla	2000	63.55%	—	—	—
BCI-E12b	2000	66.35%	+2.8 pp	7.9%	—
BCI-E12b (tuned)	2000	66.35%	+3.45 pp	7.9%	—
BCI-E1	2000	**79.75%**	**+16.2 pp**	**21.2%**	—

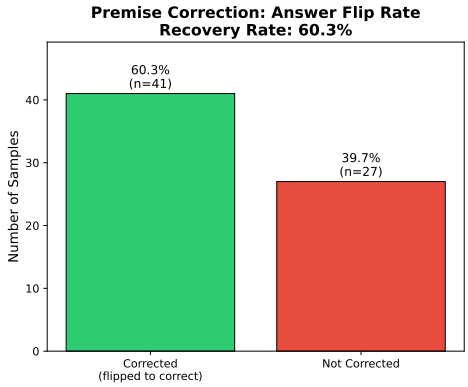

Figure 3. Premise correction recovery rate among premise-error cases (n=68). Correcting contradicted beliefs flips 60.3% (41/68) of premise-error cases to the correct answer, confirming that false visual premises are the direct causal driver in the majority of these failures.

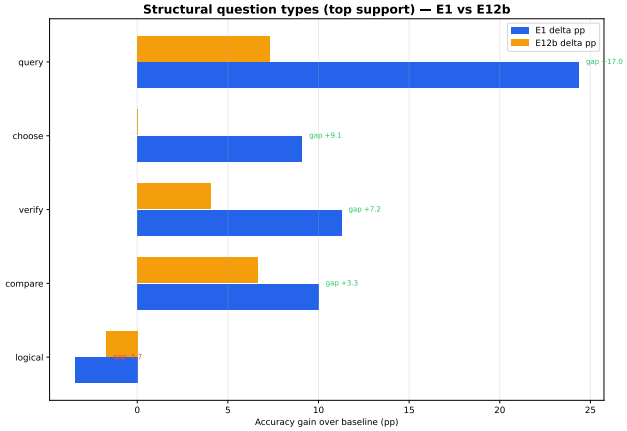

Figure 4. Accuracy gain (pp) by structural question type for E1 (blue) and E12b (orange) at n=500. E1 substantially outperforms E12b for query-type questions (+17.0 pp gap) and choose-type (+9.1 pp gap). Logical questions show a slight negative for both, reflecting the low sample count (n=116) and high baseline accuracy.

5.3. Question-Type Stratification

Table 3 breaks down E1 and E12b gains by question type at n=2000. E1 delivers the largest absolute gains for query-type questions (+25.9 pp; flip-to-correct 29.8%) and other-type questions (+20.3 pp; flip-to-correct 24.3%). Verify-type questions, which have high baseline accuracy (80.4%), show smaller but still meaningful gains (+5.6 pp). E12b shows a similar pattern but with attenuated gains, particularly for verify (+0.35 pp), reflecting the higher difficulty of identifying high-confidence contradictions in short yes/no-style questions.

Table 3. Stratified gains by question type (GQA, n=2000). E1 and E12b flip-to-correct rates reveal where premise correction is most effective.

Type	n	Base	E1 Δ	E12b Δ
query	887	47.0%	+25.9 pp	+6.2 pp
verify	854	80.4%	+5.6 pp	+0.4 pp
logical	57	75.4%	+8.8 pp	+7.0 pp
other	202	61.4%	+20.3 pp	+3.5 pp

Figures 4 and 5 further decompose these gains. Among structural question types, E1 most strongly advantages query-type (+17.0 pp gap over E12b) and choose-type (+9.1 pp gap) questions. Among semantic sub-types, categorical ("cat") questions show the largest E1 advantage (+20.97 pp gap), followed by relational questions (+17.30 pp), confirming that premise correction is especially impactful when the answer depends on identifying object categories and spatial/relational facts.

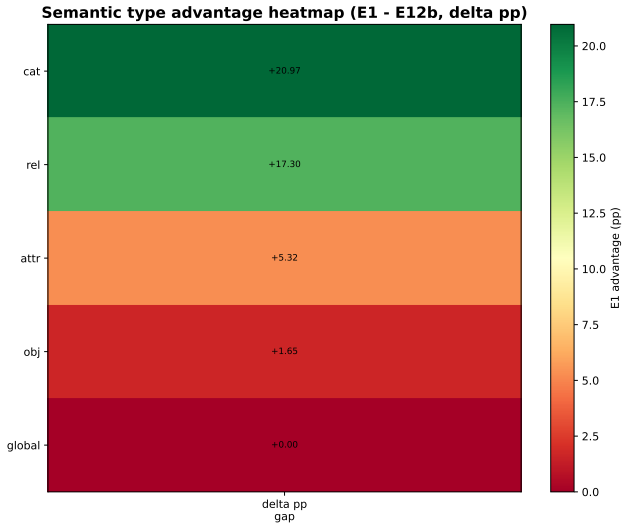

Figure 5. E1 advantage over E12b (Δpp) by semantic question sub-type (heatmap). Categorical (cat: +20.97 pp) and relational (rel: +17.30 pp) sub-types show the strongest E1 advantage, consistent with object-existence and spatial claims being the most verifiable claim families.

5.4. Targeted vs. Random Correction

To confirm that gains are claim-specific and not a consequence of generic belief perturbation, we run a *random ablation* control that removes beliefs at random using the same budget as targeted correction. As shown in Fig. 6, the random flip rate is 25.0% vs. 60.3% for targeted correction—a gap of +35.3 pp. This rules out the hypothesis that any belief modification improves VLM answers, and confirms that the mechanism is grounded in accurate identification

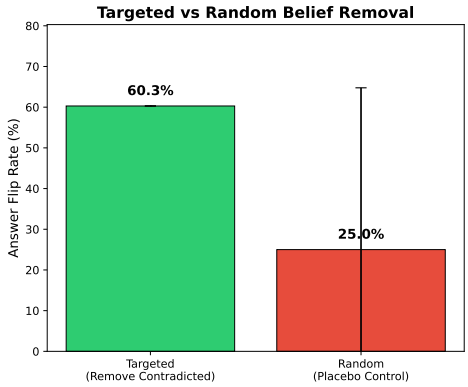

Figure 6. Targeted vs. random belief removal on premise-error cases. Targeted correction (remove only contradicted beliefs) achieves a 60.3% flip-to-correct rate versus 25.0% for random removal at matched budget, a gap of +35.3 pp confirming that the intervention is claim-specific and not a generic perturbation effect.

and replacement of false premises.

5.5. Competitor Matrix

Table 4 ranks all methods on GQA (n=500) under identical model and data conditions. BCI-E1 leads by a margin of +16.6 pp over the next-best method (Resample-Lite at 60.6%). Self-Correct underperforms Vanilla by 10.0 pp, consistent with prior findings that naive self-revision can introduce errors [8]. CoVe-Lite also underperforms Vanilla, confirming that BCI's advantage over CoVe-Lite is not attributable to the verification step alone but to the explicit premise-replacement mechanism.

Table 4. Full competitor matrix (GQA, n=500, Qwen2.5-VL-7B-Instruct). All methods share the same model and question split.

Method	Accuracy	Rank
BCI-E1	77.2% (386/500)	1
Resample-Lite	60.6% (303/500)	2
Vanilla	59.2% (296/500)	3
BCI-E3	58.6% (293/500)	4
CoVe-Lite	57.2% (286/500)	5
Self-Correct	49.2% (246/500)	6

5.6. Statistical Significance

Table 5 reports a condensed summary of pairwise McNemar exact p-values across methods. All key comparisons involving E1 are significant at $p<10^{-13}$ at n=500 and $p<10^{-47}$ at n=2000. E12b reaches significance at n=2000 (p=1.34×10^{-5}, Holm-corrected). The significance of BCI-E1 vs. E12b (p=9.99×10^{-43}) confirms that the two policies represent meaningfully different operating points on the accuracy/compute trade-off. BCI-E3 is not significant (p=0.84), providing a clean negative control.

Table 5. Pairwise McNemar exact p-values. BCI-E1 is highly significant at all scales; E12b is significant at n=2000; E3 is not significant (negative control).

Comparison	n	p-value	Sig.?
Baseline vs. BCI-E1	500	1.49×10^{-14}	✓
Baseline vs. BCI-E3	500	0.839	×
Baseline vs. BCI-E1	2000	1.78×10^{-48}	✓
Baseline vs. E12b	2000	1.34×10^{-5}	✓
BCI-E1 vs. E12b	2000	9.99×10^{-43}	✓

Figures 7 and 8 show the full pairwise significance tables at n=500 (all 56 method pairs across 6 methods) and the n=2000 subset. The complete matrix confirms that BCI-E1 wins every pairwise comparison at $p<10^{-10}$ at n=500, and that E12b is a statistically robust improvement over Vanilla at both scales while remaining significantly below E1.

6. Analysis

6.1. Verifier Audit and Robustness

Table 6 summarises verifier contradiction detection rates by claim type under the `balanced` profile (n=2000, clean vs. 20% noise injection). Across 9,780 total claims, object existence is the strongest detection family: 25.3% on clean data (844/3,337), rising to 34.7% under 20% noise injection (1,157/3,337). Attribute and spatial detection remain lower, reflecting scene-graph annotation density (2.1% and 4.5% on clean, respectively). Action claim detection is zero on clean data but rises sharply with injected noise (20.2% at 20%), suggesting that action claims are structurally verifiable but under-covered by the clean scene-graph oracle. OCR detection is effectively zero across all noise levels—a known limitation of GQA scene graphs as a verification oracle for text-in-image claims.

Table 6. Verifier contradiction detection by claim type (`balanced` profile, n=2000). Detection rates rise with injected noise; OCR is a known gap. Claim counts reflect the n=2000 run (total: 9,780 claims).

Claim Type	Count	Clean	Noise-0.1	Noise-0.2
Object Existence	3,337	25.3%	29.5%	34.7%
Attribute	2,913	2.1%	2.8%	3.8%
Action	1,502	0.0%	11.1%	20.2%
Spatial	1,604	4.2%	5.9%	7.2%
Counting	24	54.2%	50.0%	50.0%
OCR	400	0.0%	0.0%	0.0%

Profile comparison (`strict` vs. `balanced` vs. `high_recall`) at n=2000 shows near-identical behavior

on clean object detection (all 25.3%), diverging primarily on spatial and action categories under noise. This confirms that the `balanced` profile is a reliable default and that the 33.3% premise-error rate is robust to moderate verifier configuration choices.

These results have two implications for interpreting the main findings. First, the 33.3% premise-error rate is a conservative lower bound due to missed detections in attribute and action categories. Second, the verifier degrades gracefully under moderate noise (5–10%), supporting robustness of the pipeline to annotation imperfection.

6.2. Belief Externalization Quality

Table 7 compares LLaVA-1.5-7B [13] and Qwen2.5-VL-7B [2] on belief quality and downstream premise-error observability. Qwen produces 2.3× more beliefs per question on average (4.96 vs. 2.17) and achieves 14.5 pp higher baseline accuracy (59.2% vs. 52.2%). Crucially, Qwen's richer belief sets yield nearly double the observed premise-error rate (33.3% vs. 18.8%), suggesting that sparse or over-hedged beliefs suppress the observable PE rate even when premise errors are present. Our results indicate that belief externalization quality is a critical design variable: models that emit vague or entity-level beliefs rather than typed, grounded predicates limit the effectiveness of the verification stage.

Table 7. Model comparison on GQA (n=500). Richer belief externalization yields higher observed premise-error rates and better base accuracy.

Model	Acc.	PE Rate	RE Rate	Avg. Beliefs
LLaVA-1.5-7B	52.2%	18.8%	81.2%	2.17
Qwen2.5-VL-7B	59.2%	33.3%	66.7%	4.96

6.3. Belief Minimality

In 27.9% of premise-error cases (19/68), removing or replacing a *single* contradicted belief is sufficient to flip the answer to correct (see Fig. 12). This single-belief minimality rate reveals that many failures are sensitive to one false premise. The remaining 72.1% of cases likely involve multiple interacting false premises, motivating the full replacement strategy of E1 over the single-removal E3 policy. The minimality result also suggests that a future ranking of beliefs by confidence could yield efficient, surgical interventions without requiring access to full ground-truth scene graphs.

6.4. Confidence-Gated Policy Analysis (E12b)

E12b applies correction only when the intervention score $s \geq \tau$=0.18. This threshold was selected via a paired comparison against the stricter E12 variant (τ=0.25): lowering the threshold from 0.25 to 0.18 increased the intervention rate from 35.2% to 58.8%, yielding 25 additional wins at a cost of only 11 losses (McNemar p=0.0288, net +14 samples corrected; E12: 59.6%; E12b: 62.4%).

At n=2000, E12b achieves an intervention rate of 59.85% with a near-zero fallback rate of 0.10%. The resulting +3.45 pp gain shows that selective correction at roughly 60% coverage captures most of the recoverable gain under practical compute constraints. Multi-seed stability (seeds 42, 123, 456) yields $66.18\% \pm 0.14$ pp, confirming numerical robustness across random subsamples.

6.5. Cost-Efficiency

Table 8 compares E1 and E12b at n=2000 on wall-clock efficiency. E1 achieves +16.2 pp in 15,629 s (≈7.81 s/sample), while E12b achieves +3.45 pp in 15,980 s (≈7.99 s/sample). The gain-per-unit-time ratio strongly favors E1 when accuracy maximization is the goal (2.07 vs. 0.43 pp per second per sample). E12b is preferred when compute is constrained but a significant and reliable improvement is still desired.

Table 8. Cost-efficiency comparison at n=2000. E1 is superior when compute is unconstrained; E12b provides a practical alternative.

Method	s/sample	Base Acc.	Δ pp	Δ/s
E1 (n=2000)	7.81	63.55%	+16.2	2.07
E12b (n=2000)	7.99	62.90%	+3.45	0.43

Figures 13 to 16 provide a comprehensive cost-efficiency characterization across all runs.

6.6. Confidence Calibration and Abstention

We evaluate how well the intervention score s is calibrated against the observed corrected error rate, and whether score-based abstention can be used to operate BCI at different coverage/accuracy trade-offs.

Calibration. E1 achieves ECE $= 0.0793 \pm 0.0000$ across three seeds, indicating strong alignment between intervention score and error rate (Fig. 17). E12b has higher ECE $= 0.1413 \pm 0.0007$, reflecting its conditional intervention policy which skews the score distribution toward higher-confidence corrections.

Abstention. Figures 19 to 21 characterize the coverage/accuracy trade-off under score-based abstention (i.e., abstain on samples where s exceeds a threshold).

7. Conclusion

We presented **Belief-Constrained Inference** (BCI), an inference-time framework that externalizes, verifies, and corrects intermediate visual beliefs in VLMs before final answer generation. Our core finding is that premise errors

constitute 33.3% of all incorrect VLM answers on GQA, rising to 50% for logical/relational questions, and that this rate is a conservative lower bound due to verifier coverage gaps in attribute and action claim families.

Correcting these premises yields $+18.0$ pp accuracy gain at $n{=}500$ under a full-replacement oracle (McNemar $p{=}1.5{\times}10^{-14}$), scaling to $+16.2$ pp at $n{=}2000$ ($p{=}1.78{\times}10^{-48}$). A practical confidence-gated policy (E12b) achieves $+3.45$ pp at $n{=}2000$ ($p{=}1.3{\times}10^{-5}$) with 59.85% intervention rate and multi-seed stability of $66.18\% \pm 0.14$ pp. Targeted correction outperforms random belief deletion by $+35$ pp, confirming premise-specificity. Against matched lightweight baselines, BCI-E1 leads the competitor matrix by $+16.6$ pp, while Self-Correct falls below Vanilla by 10.0 pp.

Limitations. Current results are limited to GQA with deterministic scene-graph verifiers. The E1 policy is an oracle upper bound requiring ground-truth facts at test time; E12b is practical but still relies on scene-graph verification. Verifier coverage is incomplete (49% mean), especially for action and OCR claims. Cross-benchmark generalization (MMMU, MathVista, MM-Vet) and model-agnosticity (InstructBLIP, LLaVA-Next) remain for future work. Multi-seed stability for E1 at $n{=}2000$ shows zero variance across seeds, suggesting the sampling procedure may benefit from additional randomization checks.

Future Work. The most impactful extensions are: (1) replacing the scene-graph verifier with a learned or retrieval-augmented visual oracle to remove the ground-truth dependency; (2) extending evaluation to open-ended generation benchmarks; (3) incorporating uncertainty estimation into belief revision; (4) conducting a latency/accuracy Pareto analysis to characterize the full operating frontier of confidence-gated BCI; and (5) cross-benchmark generalization studies to validate premise-error patterns beyond GQA.

Broader Impact. BCI provides an interpretable audit trail—the explicit belief set and verification labels—that makes VLM reasoning more transparent and correctable. This is particularly valuable in safety-sensitive applications where understanding *why* a model failed is as important as preventing failure.

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Paired significance table: n=500 all methods (rows 1-18 of 56)

method_a	method_b	winner	n_paired	a_correct	b_correct	a_accuracy	b_accuracy	delta_accuracy_pp_a_b	a_only_correct	b_only_correct	mcnemar_exact	mcnemar_exact_p	mcnemar_chi_square
Baseline	E1	E1	500	296	386	0.592	0.772	-18.000	27	117	1.487e-14	3.421e-13	55.007
Baseline	E3	Baseline	500	296	293	0.592	0.586	0.600	50	47	8.392e-01	1.000e+00	0.041
Baseline	E12	E12	500	296	298	0.592	0.596	-0.400	23	25	8.854e-01	1.000e+00	0.021
Baseline	E12b	E12b	500	296	312	0.592	0.624	-3.200	33	49	9.703e-02	1.000e+00	2.744
Baseline	Self Correct	Baseline	500	296	246	0.592	0.492	10.000	92	42	1.871e-05	3.367e-04	17.918
Baseline	Cove Lite	Baseline	500	296	286	0.592	0.572	2.000	22	12	1.214e-01	1.000e+00	2.382
Baseline	Resample Lite	Resample Lite	500	296	303	0.592	0.606	-1.400	33	40	4.828e-01	1.000e+00	0.493
E1	Baseline	E1	500	386	296	0.772	0.592	18.000	117	27	1.487e-14	3.421e-13	55.007
E1	E3	E1	500	386	293	0.772	0.586	18.600	102	9	4.261e-21	1.110e-19	76.252
E1	E12	E1	500	386	298	0.772	0.596	17.600	106	18	2.314e-16	5.786e-15	61.040
E1	E12b	E1	500	386	312	0.772	0.624	14.800	87	13	1.313e-14	3.115e-13	53.290
E1	Self Correct	E1	500	386	246	0.772	0.492	28.000	169	29	3.146e-25	8.808e-24	97.581
E1	Cove Lite	E1	500	386	286	0.772	0.572	20.000	128	28	1.771e-16	4.606e-15	62.827
E1	Resample Lite	E1	500	386	303	0.772	0.606	16.600	117	34	7.202e-12	1.585e-10	44.530
E3	Baseline	Baseline	500	293	296	0.586	0.592	-0.600	47	50	8.392e-01	1.000e+00	0.041
E3	E1	E1	500	293	386	0.586	0.772	-18.600	9	102	4.261e-21	1.110e-19	76.252
E3	E12	E12	500	293	298	0.586	0.596	-1.000	29	34	6.147e-01	1.000e+00	0.254
E3	E12b	E12b	500	293	312	0.586	0.624	-3.800	20	39	1.834e-02	2.567e-01	5.492

Paired significance table: n=500 all methods (rows 19-36 of 56)

method_a	method_b	winner	n_paired	a_correct	b_correct	a_accuracy	b_accuracy	delta_accuracy_pp_a_b	a_only_correct	b_only_correct	mcnemar_exact	mcnemar_exact_p	mcnemar_chi_square
E3	Self Correct	E3	500	293	246	0.586	0.492	9.400	96	49	1.176e-04	1.999e-03	14.593
E3	Cove Lite	E3	500	293	286	0.586	0.572	1.400	53	46	5.467e-01	1.000e+00	0.364
E3	Resample Lite	Resample Lite	500	293	303	0.586	0.606	-2.000	49	59	3.866e-01	1.000e+00	0.750
E12	Baseline	E12	500	298	296	0.596	0.592	0.400	25	23	8.854e-01	1.000e+00	0.021
E12	E1	E1	500	298	386	0.596	0.772	-17.600	18	106	2.314e-16	5.786e-15	61.040
E12	E3	E12	500	298	293	0.596	0.586	1.000	34	29	6.147e-01	1.000e+00	0.254
E12	E12b	E12b	500	298	312	0.596	0.624	-2.800	11	25	2.882e-02	3.746e-01	4.694
E12	Self Correct	E12	500	298	246	0.596	0.492	10.400	98	46	1.755e-05	3.334e-04	18.062
E12	Cove Lite	E12	500	298	286	0.596	0.572	2.400	41	29	1.882e-01	1.000e+00	1.729
E12	Resample Lite	Resample Lite	500	298	303	0.596	0.606	-1.000	44	49	6.785e-01	1.000e+00	0.172
E12b	Baseline	E12b	500	312	296	0.624	0.592	3.200	49	33	9.703e-02	1.000e+00	2.744
E12b	E1	E1	500	312	386	0.624	0.772	-14.800	13	87	1.313e-14	3.115e-13	53.290
E12b	E3	E12b	500	312	293	0.624	0.586	3.800	39	20	1.834e-02	2.567e-01	5.492
E12b	E12	E12b	500	312	298	0.624	0.596	2.800	25	11	2.882e-02	3.746e-01	4.694
E12b	Self Correct	E12b	500	312	246	0.624	0.492	13.200	112	46	1.554e-07	3.264e-06	26.741
E12b	Cove Lite	E12b	500	312	286	0.624	0.572	5.200	60	34	9.548e-03	1.432e-01	6.649
E12b	Resample Lite	E12b	500	312	303	0.624	0.606	1.800	59	50	4.437e-01	1.000e+00	0.587
Self Correct	Baseline	Baseline	500	246	296	0.492	0.592	-10.000	42	92	1.871e-05	3.367e-04	17.918

Paired significance table: n=500 all methods (rows 37-54 of 56)

method_a	method_b	winner	n_paired	a_correct	b_correct	a_accuracy	b_accuracy	delta_accuracy_pp_a_b	a_only_correct	b_only_correct	mcnemar_exact	mcnemar_exact_p	mcnemar_chi_square
Self Correct	E1	E1	500	246	386	0.492	0.772	-28.000	29	169	3.146e-25	8.808e-24	97.581
Self Correct	E3	E3	500	246	293	0.492	0.586	-9.400	49	96	1.176e-04	1.999e-03	14.593
Self Correct	E12	E12	500	246	298	0.492	0.596	-10.400	46	98	1.755e-05	3.334e-04	18.062
Self Correct	E12b	E12b	500	246	312	0.492	0.624	-13.200	46	112	1.554e-07	3.264e-06	26.741
Self Correct	Cove Lite	Cove Lite	500	246	286	0.492	0.572	-8.000	35	75	1.724e-04	2.758e-03	13.827
Self Correct	Resample Lite	Resample Lite	500	246	303	0.492	0.606	-11.400	34	91	3.474e-07	6.948e-06	25.088
Cove Lite	Baseline	Baseline	500	286	296	0.572	0.592	-2.000	12	22	1.214e-01	1.000e+00	2.382
Cove Lite	E1	E1	500	286	386	0.572	0.772	-20.000	28	128	1.771e-16	4.606e-15	62.827
Cove Lite	E3	E3	500	286	293	0.572	0.586	-1.400	46	53	5.467e-01	1.000e+00	0.364
Cove Lite	E12	E12	500	286	298	0.572	0.596	-2.400	29	41	1.882e-01	1.000e+00	1.729
Cove Lite	E12b	E12b	500	286	312	0.572	0.624	-5.200	34	60	9.548e-03	1.432e-01	6.649
Cove Lite	Self Correct	Cove Lite	500	286	246	0.572	0.492	8.000	75	35	1.724e-04	2.758e-03	13.827
Cove Lite	Resample Lite	Resample Lite	500	286	303	0.572	0.606	-3.400	31	48	7.116e-02	8.540e-01	3.241
Resample Lite	Baseline	Resample Lite	500	303	296	0.606	0.592	1.400	40	33	4.828e-01	1.000e+00	0.493
Resample Lite	E1	E1	500	303	386	0.606	0.772	-16.600	34	117	7.202e-12	1.585e-10	44.530
Resample Lite	E3	Resample Lite	500	303	293	0.606	0.586	2.000	59	49	3.866e-01	1.000e+00	0.750
Resample Lite	E12	Resample Lite	500	303	298	0.606	0.596	1.000	49	44	6.785e-01	1.000e+00	0.172
Resample Lite	E12b	E12b	500	303	312	0.606	0.624	-1.800	50	59	4.437e-01	1.000e+00	0.587

Paired significance table: n=500 all methods (rows 55-56 of 56)

method_a	method_b	winner	n_paired	a_correct	b_correct	a_accuracy	b_accuracy	delta_accuracy_pp_a_b	a_only_correct	b_only_correct	mcnemar_exact	mcnemar_exact_p	mcnemar_chi_square
Resample Lite	Self Correct	Resample Lite	500	303	246	0.606	0.492	11.400	91	34	3.474e-07	6.948e-06	25.088
Resample Lite	Cove Lite	Resample Lite	500	303	286	0.606	0.572	3.400	48	31	7.116e-02	8.540e-01	3.241

Figure 7. Full pairwise McNemar significance table at n=500 (all 56 pairs across 6 methods: Baseline, E1, E3, E12, E12b, Self-Correct, CoVe-Lite, Resample-Lite). Cells highlighted green/red indicate significant wins/losses; yellow indicates marginal significance. BCI-E1 wins every comparison at $p<10^{-10}$.

Paired significance table: n=2000 subset (rows 1-6 of 6)

method_a	method_b	winner	n_paired	a_correct	b_correct	a_accuracy	b_accuracy	delta_accuracy_pp_a_b	a_only_correct	b_only_correct	mcnemar_exact	mcnemar_exact_p	mcnemar_chi_square
Baseline	E1	E1	2000	1271	1595	0.635	0.797	-16.200	100	424	1.780e-48	5.339e-48	199.101
Baseline	E12b	E12b	2000	1271	1327	0.635	0.663	-2.800	123	179	1.509e-03	1.509e-03	10.017
E1	Baseline	E1	2000	1595	1271	0.797	0.635	16.200	424	100	1.780e-48	5.339e-48	199.101
E1	E12b	E1	2000	1595	1327	0.797	0.663	13.400	340	72	9.988e-43	1.998e-42	173.032
E12b	Baseline	E12b	2000	1327	1271	0.663	0.635	2.800	179	123	1.509e-03	1.509e-03	10.017
E12b	E1	E1	2000	1327	1595	0.663	0.797	-13.400	72	340	9.988e-43	1.998e-42	173.032

Figure 8. Pairwise McNemar significance at n=2000 (Baseline, E1, E12b). E1 vs. Baseline: p=1.78×10^{-48}; E12b vs. Baseline: p=1.51×10^{-3} (Holm-corrected); E1 vs. E12b: p=9.99×10^{-43}.

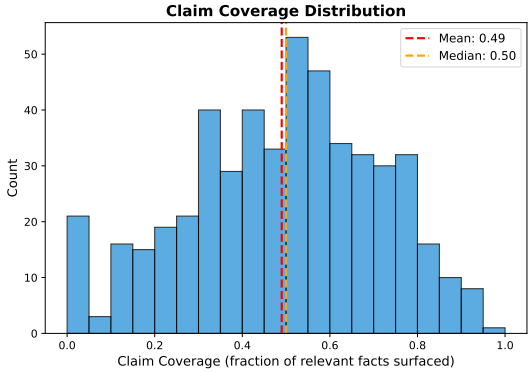

Figure 9. Distribution of claim coverage over scene-graph facts (Qwen2.5-VL, n=500). Mean coverage is 0.49, meaning roughly half of all verifiable scene facts are surfaced as explicit beliefs per question. This low coverage implies that the 33.3% premise-error rate is a conservative lower bound.

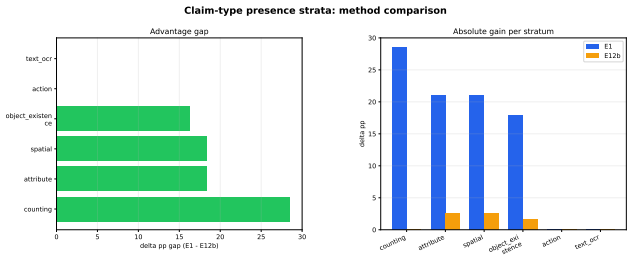

Figure 10. Claim-type presence strata: E1 and E12b accuracy gain (pp) for samples containing each claim type. Counting (+28.57 pp E1 gap), attribute (+18.42 pp), and spatial (+18.42 pp) strata show the largest E1 advantage, while action and OCR strata are zero due to verifier coverage gaps on these claim families.

Dominant contradicted type comparison

Dominant Type	n E1	n E12b	E1 Delta pp	E12b Delta pp	Gap (E1-E12b)	Corrected Acc Gap
counting	5	5	+20.00	+0.00	+20.00	+0.20
spatial	35	35	+20.00	+2.86	+17.14	+0.11
attribute	30	30	+13.33	-3.33	+16.67	+0.20
object_existence	301	301	+17.94	+1.66	+16.28	+0.19
none	653	653	+15.47	+6.74	+8.73	+0.10

Figure 11. Dominant contradicted claim type comparison (E1 vs. E12b, n=500 subset). When counting or spatial claims dominate the contradiction profile, E1 outperforms E12b by +17–20 pp. Object-existence-dominated samples show a +16.28 pp gap. Even samples with no detectable contradiction ("none") benefit from E1 (+15.47 pp), likely via implicit false beliefs missed by the verifier.

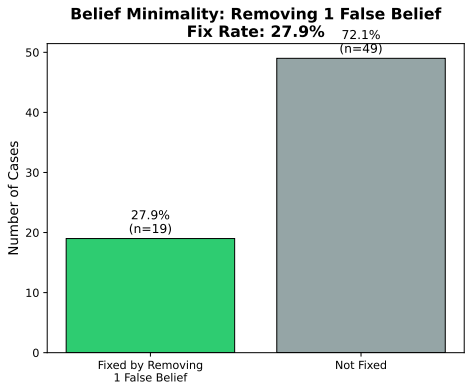

Figure 12. Belief minimality among premise-error cases (n=68). In 27.9% of cases (19/68), removing a *single* contradicted belief suffices to flip the answer to correct, revealing high sensitivity to individual false premises. The remaining 72.1% involve multiple interacting errors, motivating the full replacement strategy of E1.

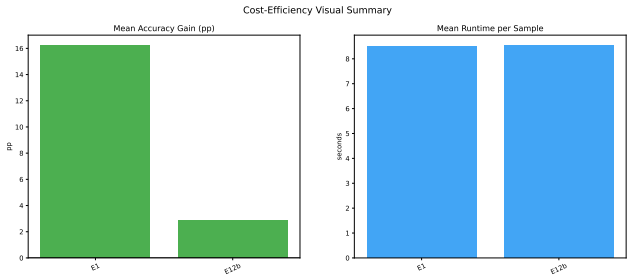

Figure 13. Cost-efficiency visual summary: mean accuracy gain (pp, left) and mean runtime per sample (seconds, right) for E1 and E12b across all runs. E1 achieves 5.8× higher mean gain (16.2 vs. 2.85 pp) at essentially identical per-sample cost (8.50 vs. 8.53 s/sample).

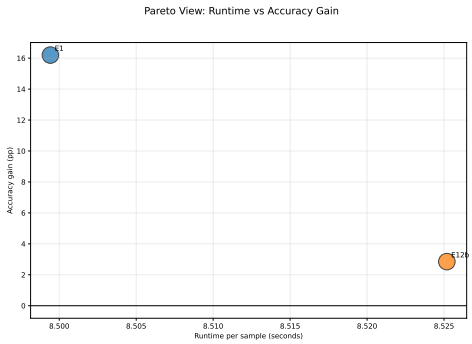

Figure 14. Pareto view: runtime per sample vs. accuracy gain. E1 occupies the dominant position (top-left), achieving +16.2 pp at 8.50 s/sample. E12b offers a useful fallback at +2.85 pp for lower-certainty deployments.

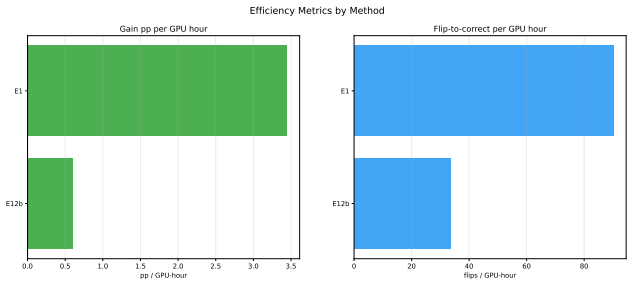

Figure 15. GPU-hour efficiency: gain pp/GPU-hour (left) and flips/GPU-hour (right). E1 achieves 3.44 pp/GPU-hour and 90 flips/GPU-hour vs. E12b at 0.61 pp/GPU-hour and 33 flips/GPU-hour, confirming E1's dominance when accuracy per unit compute is the objective.

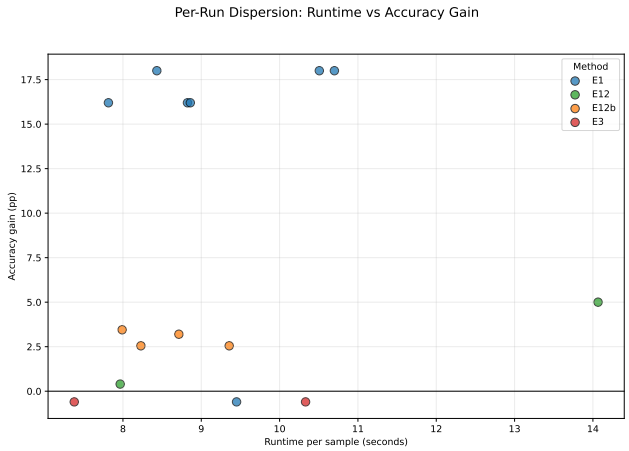

Figure 16. Per-run dispersion: runtime vs. accuracy gain across all 18 individual runs (E1 blue, E12b orange, E12 green, E3 red). E1 runs cluster at +16–18 pp gain with 7.8–10.7 s/sample; E12b clusters at +2.5–3.5 pp; E3 runs show near-zero or negative gain, confirming that removal-only is not a viable policy.

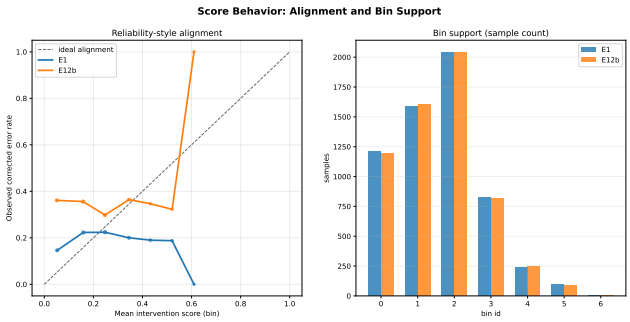

Figure 17. Reliability diagram and bin support for E1 (blue) and E12b (orange). Left: E1 tracks the ideal diagonal closely (ECE$=0.079$); E12b deviates more at high scores (ECE$=0.141$). Right: bin support shows most samples fall in score bins 1–2 ($s \in [0.1, 0.3]$), indicating the majority of interventions operate in a well-calibrated mid-confidence regime.

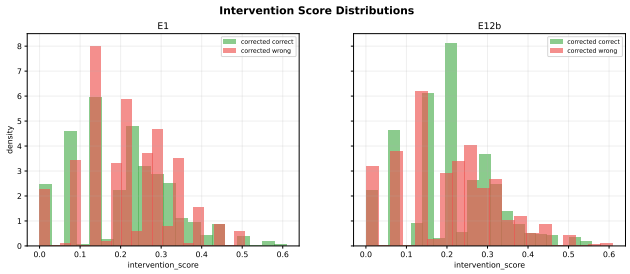

Figure 18. Intervention score distributions for E1 (left) and E12b (right), stratified by outcome (corrected-correct: green; corrected-wrong: red). E1 shows substantial overlap between distributions, reflecting its unconditional policy. E12b's gating concentrates corrections in the $s \in [0.15, 0.30]$ range where green outcomes dominate.

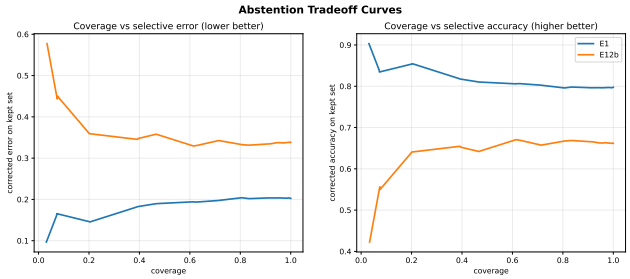

Figure 19. Abstention trade-off curves. Left: as coverage decreases (more abstentions), E1 achieves lower corrected error (reaching $\sim$10% at 5% coverage), while E12b plateaus near 33%. Right: E1 selective accuracy peaks at $\sim$90% at low coverage, confirming that E1's high-confidence corrections are very reliable.

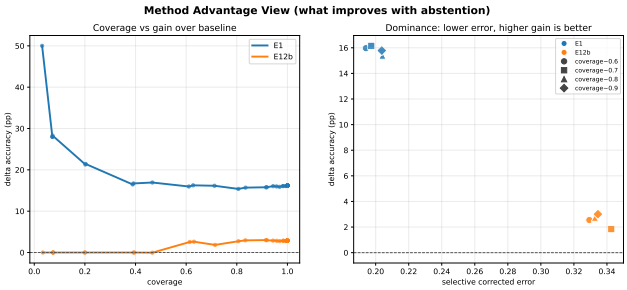

Figure 20. Left: coverage vs. gain over baseline (Δpp). E1 maintains $+16$ pp gain across all coverage levels above 40%, demonstrating broad-coverage robustness. E12b gain varies between 0–3 pp across coverage. Right: dominance scatter (error vs. gain) at four coverage targets (0.6–0.9); E1 operating points cluster tightly near (0.20, $+16$ pp), confirming stable Pareto dominance over E12b.

Operating Points (coverage targets)

method	operating_point	coverage	threshold	corrected_accuracy	corrected_error_rate	delta_accuracy_pp_k	value
E1	AURC_like						0.1762
E1	coverage−0.6	0.611	0.225	0.806	0.194	15.97	
E1	coverage−0.7	0.712	0.275	0.803	0.197	16.15	
E1	coverage−0.8	0.806	0.300	0.796	0.204	15.38	
E1	coverage−0.9	0.916	0.350	0.796	0.204	15.78	
E12b	AURC_like						0.3458
E12b	coverage−0.6	0.614	0.225	0.671	0.329	2.55	
E12b	coverage−0.7	0.714	0.275	0.657	0.343	1.84	
E12b	coverage−0.8	0.807	0.300	0.667	0.333	2.73	
E12b	coverage−0.9	0.915	0.350	0.665	0.335	3.01	

Figure 21. Operating points table (coverage targets 0.6–0.9). At 60% coverage (τ=0.225), E1 achieves 80.6% selective accuracy ($+16.0$ pp over baseline); at 90% coverage (τ=0.350), E1 maintains 79.6% ($+15.8$ pp). E12b selective accuracy ranges from 65.7%–67.1% across the same thresholds, confirming that even selective E12b cannot close the gap to E1.